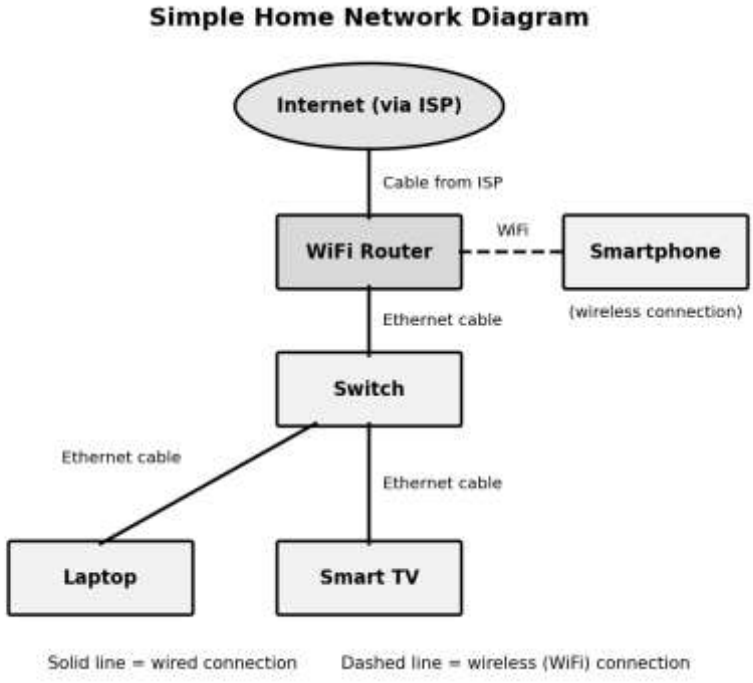
Network Devices

Q1. Draw and label a simple home network diagram showing a router, switch, and three devices (like a laptop, smartphone, and smart TV), indicating how each device connects to the network.



Part	How it connects	Role
WiFi Router	Cable from the ISP on one side; cable to the switch and WiFi to the phone on the other side	Connects the home network to the internet and gives IP addresses to the devices.
Switch	Ethernet cable to the router	Gives extra wired ports and sends data only to the device that needs it.
Laptop	Wired, Ethernet cable to the switch	Stable and fast connection for work or video calls.
Smart TV	Wired, Ethernet cable to the switch	Smooth streaming of videos (for example YouTube) without buffering.
Smartphone	Wireless, WiFi from the router	Moves freely around the house while staying connected.

Q2. Create a comparison table listing at least four network devices (router, switch, hub, access point) and describe one real-world use case for each, referencing apps or services you use daily (e.g., WiFi for streaming on YouTube, LAN gaming, etc.).

Device	What it does	Real-world use case
Router	Connects different networks (home network and the internet) and finds the path for data using IP addresses. It also gives IP addresses to devices (DHCP).	The home WiFi router that lets the phone, TV and laptop open YouTube, Instagram and Zomato by connecting the home network to the ISP.

Device	What it does	Real-world use case
Switch	Connects many wired devices in one LAN. It learns each device's MAC address and sends data only to the correct port.	A college computer lab or an LAN gaming zone, where many PCs are connected through a switch so they can play multiplayer games with low delay.
Hub	Connects many wired devices, but it simply copies the data and sends it out of all ports. It has no intelligence, so it is slow and less secure.	Used in old, small networks (for example older cyber cafes or training labs) for basic file sharing and printer sharing. It is now replaced by switches.
Access Point (AP)	Gives wireless (WiFi) access to a wired network. It connects WiFi devices to the wired LAN.	The WiFi in a college campus, mall or cafe, so phones can stream YouTube videos and open Instagram without any cables.

Q3. Write a short scenario where you explain how data travels from your phone to the internet when you order food on Zomato, mentioning the role of each network device involved (router, switch, access point).

Scenario: I am at home (or at college) and I tap "Place Order" in the Zomato app on my phone. The following happens in a few moments:

1. Phone to Access Point: My phone sends the order details as a wireless (WiFi) signal. The access point receives it and passes it to the wired network. In a home router, the access point is built into the router.
2. Access Point to Switch: In a bigger network (for example college), the access point is connected to a switch. The switch looks at the MAC address in the data and sends it only to the correct port, which leads to the router.
3. Switch to Router: The router receives the data. It checks the destination IP address (Zomato's server), and changes my private IP address to the public IP address of my connection (NAT).
4. Router to ISP and Internet: The router forwards the request to the ISP (such as Jio or Airtel), which sends it through many routers on the internet to Zomato's server.
5. Zomato server: The server saves the order, sends it to the restaurant and sends back a confirmation.
6. Reply back: The confirmation travels back along the same path: internet, ISP, router, switch, access point, and finally my phone, where the app shows "Order placed".

Role of each device

Device	Role in this scenario
Access Point	Receives the phone's WiFi signal and connects the phone to the wired network.
Switch	Forwards the data inside the local network only to the correct device (router), which keeps the network fast.
Router	Sends the data out of the local network to the internet and brings the reply back to my phone.

Q4Given this situation: you have a WiFi router, but your gaming PC only has an Ethernet port and is far from the router. Suggest two different network devices you could use to connect your PC to the internet and explain how each would work in this setup. Think about devices like switches, powerline adapters, or WiFi extender

Option 1: Powerline adapter kit

- A kit has two adapters. The first adapter is plugged into a power socket near the router and connected to the router's LAN port with an Ethernet cable.

- The second adapter is plugged into a power socket near the gaming PC, and the PC is connected to it with a short Ethernet cable.
- The adapters send the network data through the existing electrical wiring of the house, so no new long cable is needed. It gives a wired-like connection and is more stable than weak WiFi.
- Tips: Plug the adapters directly in the wall (not in a power strip), and both must be on the same electrical circuit. Speed depends on the quality of the house wiring.

Option 2: WiFi extender (or WiFi-to-Ethernet adapter)

- A WiFi extender that has an Ethernet port is placed midway, where the router's WiFi signal is still strong. It receives the signal from the router and repeats it.
- The gaming PC is connected to the extender's Ethernet port with a cable (or with a WiFi-to-Ethernet adapter), so the PC joins the home network through WiFi even though it has no WiFi card.
- Tips: Place it halfway between the router and the PC. Extenders can reduce speed because they use the same WiFi channel to receive and send, so a dual-band or mesh extender is better.

Which one is better?

Point	**Powerline adapter**	**WiFi extender**
Connection to router	Through electrical wiring	Through WiFi signal
Speed and delay	Usually more stable for gaming if the wiring is good	May be slower with higher delay because of WiFi sharing
Setup	Plug in two adapters and press the pair button	Place midway and connect to the router's WiFi
Best when	The house wiring is in good condition	The WiFi signal is strong enough in the middle spot

Conclusion: For gaming, the powerline adapter is usually the better choice because it gives a stable wired link; a WiFi extender is a good second option when powerline does not work well. If a long cable is possible, running a Cat6 cable to a switch near the PC is the most reliable option.